60M (0.06B) best pretrain LR by batch size and token budget
cell = best LR / its DCLM_heldout loss; row = chinchilla; wd 0.1

	adamw bs 1M	adamw bs 2M	adamw bs 4M	muon bs 1M	muon bs 2M	muon bs 4M
0.25	0.056 5.281	0.11 6.084	0.16 6.628	0.014 4.505	0.0099 5.132	0.014 6.033
0.5	0.04 4.540	0.079 5.238	0.16 5.987	0.01 4.180	0.014 4.427	0.014 5.079
1	0.02 4.167	0.04 4.432	0.056 5.145	0.014 3.999	0.014 4.099	0.014 4.340
2	0.02 3.986	0.02 4.083	0.028 4.367	0.01 3.916	0.014 3.942	0.014 4.047
4	0.01 3.885	0.014 3.938	0.014 4.073	0.014 3.855	0.0099 3.877	0.014 3.909
8	0.014 3.832	0.014 3.861	0.02 3.906	0.007 3.807	0.0071 3.820	0.02 3.833
16	0.01 3.798	--	--	0.014 3.780	--	--
32	0.01 3.769	--	--	0.007 3.755	--	--
64	0.007 3.748	--	--	0.005 3.743	--	--
128	0.01 3.728	--	--	0.005 3.723	--	--